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Lecture due Jan 24, 2024 20:30 IST

A system of 3 equations with 3 unknowns,

$$8y - 4z = 0$$

$$x - y - 4z = 1$$

$$-x + 5y + 2z = 0,$$

can be written in matrix form as $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 0 & 8 & -4 \\ 1 & -1 & -4 \\ -1 & 5 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$\mathbf{A}$ $\mathbf{x}$ $\mathbf{b}$

We can think of multiplying the vector $\mathbf{x}$ and the matrix $\mathbf{A}$ in 2 equivalent ways as follows:

1. To multiply $\mathbf{A}$ and $\mathbf{x}$ to get a vector $\mathbf{b}$, the i th entry of $\mathbf{b}$ is the dot product of the i th row of $\mathbf{A}$ with $\mathbf{x}$.

$$\begin{pmatrix} 0 & 8 & -4 \\ 1 & -1 & -4 \\ -1 & 5 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8y - 4z \\ x - y - 4z \\ -x + 5y + 2z \end{pmatrix}$$

This explains why the matrix form is equivalent to the original system of equations.

2. The product $\mathbf{Ax}$ can be expressed as a linear combination of the columns of $\mathbf{A}$:

$$\begin{pmatrix} 0 & 8 & -4 \\ 1 & -1 & -4 \\ -1 & 5 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + y \begin{pmatrix} 8 \\ -1 \\ 5 \end{pmatrix} + z \begin{pmatrix} -4 \\ -4 \\ 2 \end{pmatrix}.$$

Notice that the coefficients x , y , and z in this linear combination are the entries of the vector $\mathbf{x}$.

Matrix vector multiplication, dot product with rows

1/1 point (graded)

Find the missing 3rd entry c of the vector given by the matrix vector product

$$\begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ c \\ -1 \end{pmatrix}$$

$c =$

✓ Answer: 0

Solution:

To find the 3rd entry, we need only find the dot product of the given vector with the 3rd row of the given matrix. This is

$0(1) + 1(2) - 2(3) + 1(4) = 0.$

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i Answers are displayed within the problem

Matrix vector multiplication, combinations of columns viewpoint

3/3 points (graded)

Find a nonzero vector $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ such that $\mathbf{Ax} = \mathbf{0}$ where $\mathbf{A}$ is the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 3 \\ -2 & -8 & 1 \\ 0 & 0 & 4 \\ 1 & 4 & 2 \end{pmatrix}$$

(Hint: the second column is 4 times the first column.)

$x =$ ✓ Answer: 4

$y =$ ✓ Answer: -1

$z =$ ✓ Answer: 0

Solution:

When we think of matrix-vector multiplication as taking a linear combination of the columns of the matrix $\mathbf{A}$, we can rephrase this question as asking for a linear combination of the columns of $\mathbf{A}$ that give the zero vector. Note that this is only possible if the columns of $\mathbf{A}$ are linearly dependent. By inspection, we see that the second column is 4 times the first column, thus there are linearly dependent. The third column is independent since it has a nonzero 3rd

entry. Therefore one possible answer is given by $\mathbf{x} = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}$ which gives us

$$\mathbf{Ax} = \begin{pmatrix} 1 & 4 & 3 \\ -2 & -8 & 1 \\ 0 & 0 & 4 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 4 \\ -8 \\ 0 \\ 4 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

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4. Writing systems in matrix form

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? after solving the above question, the answer I'm getting is infinite solution for x and y.

4

$x+4y+3z=0$ $-2x-8y+z=0$ $z=0$ $x+4y+2z=0$ after solving the equations no unique solution was obtained for the...

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